

AI/ML Task 3 — Methodology & Results Report

Model Validation, Overfitting Control & Hyperparameter Tuning | Maincrafts Technology

1. Objective

This report documents the Task 3 pipeline, which extends the Task 2 House Price Prediction system by introducing model validation through cross-validation, overfitting detection, and systematic hyperparameter tuning using GridSearchCV. The focus shifts from raw accuracy to model reliability and generalization, a critical distinction in production ML systems.

2. Dataset

The same California Housing Dataset (20,640 samples, 8 features) from Task 2 was retained to ensure fair comparison. Feature scaling via StandardScaler and an 80/20 train-test split (random_state=42) were applied identically.

3. Overfitting Analysis

An unconstrained Decision Tree (no depth limit) was trained first to deliberately demonstrate overfitting. The results were stark:

Metric	Untuned Decision Tree	Tuned Decision Tree
Train RMSE	0.0	0.2977
Test RMSE	0.4188	0.3212
Overfit Gap	0.4188	0.0235

The untuned model memorized training data perfectly (Train RMSE = 0.0000) but generalized poorly. This massive gap of 0.4188 between train and test error is a textbook example of overfitting. Hyperparameter constraints (max_depth, min_samples_split) were used to correct this.

4. Cross-Validation

5-Fold Cross-Validation was applied to obtain a robust performance estimate independent of any single train-test partition. Rather than evaluating on one fixed split, the model is trained and evaluated 5 times on different data subsets, and scores are averaged.

CV RMSE (Untuned DT): 0.4178 +/- 0.0046

The low standard deviation (+/- 0.004) confirms the model behaves consistently across folds, meaning the CV estimate is trustworthy and not a lucky/unlucky split artifact.

5. Hyperparameter Tuning via GridSearchCV

GridSearchCV performed an exhaustive search over 12 combinations (4 max_depth values x 3 min_samples_split values) using 5-fold CV, selecting the combination with the lowest average RMSE.

Parameter	Values Tested	Best Value
max_depth	[3, 5, 7, 10]	7
min_samples_split	[2, 5, 10]	10

Best CV RMSE achieved: **0.3187**

6. Final Model Comparison

Performance of all three models on the held-out test set:

Model	Train RMSE	Test RMSE	R² Score	CV RMSE	Rank
Linear Regression	0.3944	0.3944	0.9048	0.3923	3rd
Ridge Regression	0.3944	0.3944	0.9048	0.3923	2nd
Tuned Decision Tree	0.2977	0.3212	0.9368	0.3187	1st

7. Conclusion & Final Model Justification

The Tuned Decision Tree (max_depth=7, min_samples_split=10) was selected as the final model based on four key criteria:

- Lowest Test RMSE** (0.3212) — most accurate on unseen data.
- Highest R² Score** (0.9368) — explains 93.7% of house price variance.
- Best CV RMSE** (0.3187) — performance is consistent across all 5 folds.
- Controlled overfitting** — overfit gap reduced from 0.4188 (untuned) to 0.0235 (tuned).

Linear and Ridge Regression remain interpretable alternatives but underfit the non-linear patterns in housing data. The tuned tree strikes the right bias-variance balance, making it production-ready. The final model was persisted using joblib for deployment.